

CLASS-12

SENIOR SECONDARY EXAMINATION -2016

BIOLOGY

TIME: 3 ¼ HOURS

MM: 56

GENERAL INSTRUCTION TO THE EXAMINEES.

- 1. Candidates must write first his/ her roll number on the question paper compulsorily.**
- 2. All questions are Compulsory.**

Write the answer to each question in the given answer book only .

For questions having more than one part the answers to those parts are to be written together in continuity.

If there is any error difference contradiction in Hindi and English versions of the question paper the questions of Hindi version should be treated valid.

For questions having more than one part the answers to the other part should be written together in continuity.

1. Section	q.no	marks
A	1-8	1
B	9-14	2
C	15-24	3
D	25-30	5

- 2. There are internal choices in question 27 to 30.**

SECTION-A

- 1) What is emasculation? [1]**

Ans: Emasculation is the process of removing anthers from bisexual flowers. **Write down the names of hormone secreted by ovary in the later phase of pregnancy .**

Ans: . Hormone progesterone is produced by Corpus luteum during the later phase of pregnancy it helps in maintaining pregnancy.

- 2) Mention the application of in vitro fertilization.**

Ans: In vitro fertilization is a method of a Assisted reproductive technology to treat infertility .In vitro fertilization occurs in vitro literally in glass.

3) **Write down the name of any one natural method of contraceptive.**

Ans: The names of hormone secreted by ovary in the later phase of pregnancy are following.

- Periodic abstinence
- Withdrawal method
- Lactational amenorrhea.

4) **In which continents Homo sapiens arose?**

Ans: The first homosapiens from the continent of Africa.

5) **Write down the name of typhoid causative pathogen.**

Ans: Typhoid is an infectious disease caused by bacteria Salmonella typhi.

6) **How does opioids affect the body?**

Ans: Opioids are the drugs which binds to specific opioid receptor present in our central nervous system and suppress its functioning.

7) **What is blue- Revolution?**

Ans: Blue revolution refers to enhance fish production to meet the Global food requirement.

8) **Write down the two beneficial roles of bacteria converting milk into curd.**

Ans: Lactobacillus is the bacteria which are used to convert milk into curd. These bacteria play two beneficial role .First is there add nutritional quantity by increasing vitamin B12 and also check the disease causing microbes in stomach.

9) **Define antibiotics.**

Ans: Antibiotics are the chemical substances produced by microbes that suppress the growth of another microbes.

10) **Define gene therapy .**

Ans: Gene therapy is collection of methods which allow correction of gene defect that has been diagnosed in individual.

11) **Which bacteria produces BT toxin?**

Ans: Bacteria Thuringiensis produces BT toxin.

12) Define population.

Ans: A group of individuals present in given area which can interbreed and can produce fertile offspring is called population.

SECTION – B

13) How does bisexual and unisexual conditions in fungi in plants are denoted?

Ans: In fungi bisexual are denoted by Heterothallic unisexual.

In plants bisexual are denoted by dioecious and unisexual are denoted by Monoecious.

14) Explain the process of spermatogenesis in detail.

Ans: Spermatogenesis is a process of production of sperms from a immature germ cell in male. It takes place in seminiferous tubules present inside the testis. During spermatogenesis a diploid spermatogonium male germ cell increases its size to form a diploid primary spermatocyte. This primary spermatocyte undergoes first meiotic division which is a reductional division to form two haploid equal secondary spermatocytes. Each secondary spermatocyte undergoes second meiotic division to form two equal haploids. Hence a diploid spermatogonium produces four haploid spermatids. These spermatids are transformed into spermatozoa by the process of spermiogenesis.

15) What is Co- dominance? With the help of example explain co-dominance in human being.

Ans: When two alleles of a trait equally express with neither recessive nor dominant it creates co-dominance. ABO blood group system is a good example of co-dominance. The gene 'I' for the blood group contains three alleles I^A , I^B and I which represent the phenomena of multiple allelism. as it has more than two alleles governing the same character. The alleles I^A and I^B are co-dominant. They express themselves even in the presence of the other. Thus the presence of both the alleles I^A and I^B gives the blood group of AB.

16) Write down the four salient features of genetic code.

Ans: The genetic code possesses following important property.

- The genetic code is a triplet- As singlet and triplet codes are not enough to code for 20 amino acids the minimum requirement is a triplet code.

- The genetic code is universal- Universal genetic code means that all living systems used nucleic acid and the same three base codons who direct the synthesis of protein from amino acids.
- The genetic code is not overlapping - A non overlapping codon means that the same letter is not used for two different codons. For instance, the nucleotide sequence GUU GUC represent only two codons GUU and GUC.
- The genetic code is non ambiguous-It means that a particular codon will always code for the same amino acids there wherever it is found.

17) Prove the organic evolution on the basis of comparative anatomy and morphology.

Ans: Comparative morphology and anatomy of individual organs and organ system of diverse forms of organisms reveal their phylogenetic relationship and thus provides an important evidence of evolution .

- Organs different in function but of similar embryonic origin and development have similar relationships with adjacent organ are known as homologous organ for instance arms of man, leg of horse, wing of a bird or a bat, flipper of a seal and forelimb of a cheetah are apparently different from one another in structure and function yet all of them are built on one basic plan, the pentadactyl plan.
- Analogy and analogous organs -analogous organs are those which perform the same function has superficial resemblance to one another but are of different origin and this phenomena is called analogy for example Wings of insects, Birds and bats performs the same function of flying and possess superficial resemblance but their basic structure is entirely different,
- vestigial organs are those organs which are non - functional in their present but were functional in their ancestors and in related animals.

18) What is innate immunity? How many types of barriers are present in Innate immunity .write down their names.

Ans: Innate immunity-This type of immunity is hereditary and It passes from one generation to next. It is present at the time of birth .Different types of are provided to the entry of foreign agents in our body these are

- Physical barriers - Skin in our body is the main barrier which prevents entry of microorganisms.
- Physiological barriers- acid in the stomach, saliva in the mouth, tears from eye all prevents microbial growth .
- Cellular barriers-certain types of leucocytes like polymorpho-nuclear leukocytes and monocytes in the blood and macrophages in the tissues can phagocytose and destroy microorganism.
- Cytokine barrier – virus infected cells secrete proteins called Interferons which protects non infected cells from further viral infection.

19) Write down the objectives of plant breeding. Mention the main steps in breeding any two genetic crops.

Ans: Plant breeding is the purposeful manipulation of plant species in order to create desired plants types that is better suited for cultivation, give better yield and disease resistance plant.

The objectives of plant breeding are

- Higher Yield.
- Improve quality.
- Disease controlled by developing pest resistance varieties.
- Developing varieties resistant to cold, frost and salinity .
- Developing early crop maturity to cut the cost of production.
- To develop specific agronomic characteristics such as tallness, Dwarfness, tillering .

20) Write down the applications of bio fertilizers .Explain the role of cyanobacteria as a bio fertilizer .

Ans: Bio fertilizers are substances that make use of microorganism to fertile the soil. These fertilizers are not harmful to crops or other plants like the chemical fertilizers. They are also environment friendly and do not cause the pollution of any sort. Use of bio fertilizers in the soil make the plant healthy as well as protect them from getting any diseases .

Role of Cyanobacteria as bio fertilizer- cyanobacteria can fix atmospheric nitrogen thus they increase the fertility of soil. They also add organic matter to th fertility of soil.

21) Define genetically modified organism .How GM plants are useful for us?

Ans:- An organism in which a foreign gene has been artificially transferred is known as transgenic organism .A transgenic organism is also called genetically modified organism. Several transgenic plants have been produced which are resistant to herbicides, insect, virus and host of others stresses. Transgenic plantswhich are suitable for food processing have also been evolved. Transgenic plants are used that as bioreactor of factories for producing specialtychemicals and pharmaceuticals. They have also been used for identification of regulatory sequences of many genes ,using gene construct with overlapping deletions.Transgenic plants are used for crop improvement.

22) What is biodiversity? Mention the causes of loss of biodiversity.

Ans:Biodiversity may be defined as the sum total of species richness that is the number of species of plants and animals and microorganism in a given habitat .Biodiversity is variability among living organism from all sources including internal via Terrestrial Marine and other aquatic ecosystems.

The loss of biological diversity is a global crisis. Some important causes of biodiversity losses are as follows –

- Habitat loss and fragmentation.
- Over exploitation.
- Alien species invasions.
- Co- extinction

23) What is air pollution? Describe the efforts made for controlling regular air pollution.

Ans: Air pollution can be defined as the imbalance in the quality of air so as to cause ill effect. Different types of pollutants are continuously introduced into the atmosphere and removed by self-purification process of air. When the rate of the self-purified capacity decreases it causes serious public health problems.

Pollution check by –

- Careful planning and siting of industries and better designing and operation of equipment.
- Air pollution by emissions from vehicle exhaust can be checked by using (i) correct mixture of fuel gas (ii) Additives to improve combustion (iii) by upgrading of engineering design and to install abatement equipment to improve combustion with the existing engine designs.

OR

What is pollution? Describe the study and efforts made for the remedy of plastic waste.

Ans: Pollution can be defined as adding to the environment a potentially hazardous substance or source of energy faster than the environment can accommodate it. An undesirable change in physical, chemical or biological conditions of our land and water that may have harmful effect upon human life or on undesirable species or waste that may waste or deteriorate our raw material resources. It is potentially harmful and interferes with man's use of his environment.

Plastic sack manufacturer Ahmed Khan of Bangalore developed polyblend fine powder of recycled modified plastics. This mixture is mixed with the bitumen which is used to lay roads. These roads are more water repellent and more durable.

SECTION – C

24) Write down the names of post fertilization events in flowering plants. Describe the process of embryo development.

Ans: Post fertilization events in flowering plants are followings-

- Process of embryo development-Angiosperms zygote undergoes in resting phase. When the endosperm is formed development of zygote starts. In the beginning it absorbs food from the endosperm and increases in size then after a layer is secreted by itself. Now it is called oospore. The first division of oospore is transverse wherein two cells are formed. One cell lies towards micropyle is called basal cell or suspensor cell and the other cell is formed towards Chalazal end is called apical cell or terminal cell. The basal cell and embryonal cell divide simultaneously.

The basal cell divides transversally and apical cell divides vertically resulting two suspensor cells and two embryonal cells. The two suspensor cells divide it by the transverse division forming a 6-celled long filament like structure called suspensor. The cell of suspensor lies near the embryonal cell. This cell combined with radical to form the apex of root. This four cell quadrants embryo further divide transversely to produce 8 celled stage of embryo which is termed octant stage. All the cells of octant are divided by periclinal division so that a 16 cell globular embryo is formed. Due to the fast division of embryonal cells of globular embryo, a heart shape embryo is formed.

They grow into two cotyledons. Both the growing cotyledons curve in downwards due to the curved position of body of ovule of capsella. The tissue present below the joining place of both the cotyledons are responsible to form plumule and behind it epicotyl is formed. The tissue present opposite to the plumule near the hypophysis give rise to radical. The apex root is formed by hypophysis. When embryo is formed suspensor dries and degenerates so it is known as meroblastic development. Ovule modified into seed in which testa is formed by outer integument and tegmen is formed by inner integument. Only micropyle of ovule remains unchanged and also present in seed. Ovary modified into fruit.

26. Describe the dihybrid cross experiment performed by Mendel. Explain the law of postulated by this experiment.

Ans: Mendel investigated for crosses in which the parents differed in two pairs. Such a cross which involves two pairs of contrasting characters simultaneously are called dihybrid cross. For a dihybrid experiment, Mendel crossed two parent pea plants, one of which was homozygous for round and yellow seed (RRYY) and the other for wrinkled and green seeds. Round shape and yellow colour of seed is dominant over wrinkled shape and green colour. During gamete formation each pair of alleles segregate independently of each other and produced only RY or ry gametes. All the plants produced in F₁ generation were with round and yellow seeds which were heterozygous for both the alleles and are called dihybrid. Then F₁ progeny was self pollinated this resulted in four

different combination of seeds in F2 generation their wrinkled yellow round yellow wrinkled green seeds and round green in the phenotypic ratio of 9 : 3: 3: 1.



YyRr F1 Generation Phenotype

F2 generation

	RY	Ry	rY	ry	
RY	RRYY	RRYy	RrYY	RrYy	Round/Yellow: 9
Ry	RRYy	RRyy	RrYy	Rryy	Round/green: 3
rY	RrYY	RrYy	rrYY	rrYy	wrinkled/Yellow: 3
ry	RrYy	Rryy	rrYy	rryy	wrinkled/green: 1 9:3:3:1 phenotypic ratio

Conclusion- Law of Independent Assortment-Mendel's law of independent assortment states that the alleles of two or more different genes get sorted into gametes independently of another.

27) Describe the responses found in organism against the environmental abiotic factors.(any three).

Ans:Organisms show four means to regulate their internal body environment with changing external environment:

- Regulate -Some organism maintains homeostasis by physiological means which ensure constant body temperature, constant osmotic concentration, etc. All birds and mammals are capable of such regulation. We maintain a constant body temperature of 37 degree centigrade.
- Confirm - Majority of animals and nearly all plants cannot maintain a constant internal environment. Their body temperature changes with the ambient temperature. Similarly in

aquatic animals osmotic concentration of the body fluid changes with that of ambient water osmotic concentration. These organisms are called consumers.

- Migrate -Stressful external conditions are localized or remain only for a short duration. The organism can move away temporary from the stressful Habitat to a more hospitable area and return when stressful period is over. For example thousands of migratory birds from Siberia migrate to Keolado National Park in Rajasthan every winter.

Or

Write down the adaptation found in organism to face the extreme conditions of environment factors(any three). [3 × 1 =3]

Ans: Adaptation in plants –

Hydrophytes-

In aquatic habitat hydrophytes are characterized by some special adaptation in morphological and anatomical features. Such plants have either poorly developed or completely absent. Leaves are long slender, spongy, flexible for example hydrilla. Floating leaves are large flat and entire and their upper surface is coated with wax. for example-Nelumbo.

Xerophytes- Some important morphological adaptations in this plants include reduction of leaves to small scale structure ,shedding of leaves under condition of deficient water supply ,modification of leaves into spines and such other structures ,development of water storage tissue in the leaves.

Adaptation in animals-

Adaptation in animals to face extreme conditions is following-

- In polar sea aquatic mammals like seals they have a thick layer of fat below their skin that act as insulator and reduce the loss of body heat.
- Bats, birds and insects are well adapted for an active flight they have boat or spindle shaped body that offers almost no resistance in preparing through the air forelimbs are modified into wings. Presence of well-developed flight muscles. A much convoluted and large cerebellum accounts for ability to maintain equilibrium and coordination.
- Adaptations in arboreal animals help them in climbing and are found in various animals like squirrel, insect and lizards. Their body countour is stout with his sub circular thorax and much curved ribs. The Limb Girdles are very stout giving support to body in climbing. Well-developed claws help in grasping and climbing.

SECTION- D

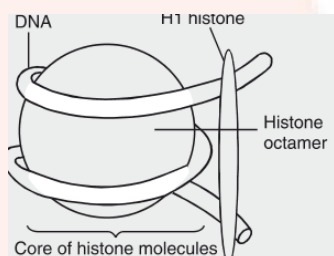
28) What is nucleosome? Explain the packaging of DNA helix. Draw the level diagram of nucleosome.

Ans: Each nucleosome consists of a spiral of DNA wrapped around an octamer of histone molecules forming core particles. The octamer of proteins consists of two molecules each of four different histones. These histones are H2A, H2B, H3 and H4.

Packaging of DNA helix in prokaryotes –

Prokaryotes lack a well-defined nucleus therefore DNA carrying negative charge is held together with some proteins that have net positive charge, in a region known as nucleoid. The DNA in nucleoid is organized in nucleus in large loops held by protein.

The organization of DNA is much more complex in eukaryotes. Depending upon the abundance of amino acid residue with charged side chains a protein acquires its charges. The histones are the proteins that are rich in lysine and arginine residue. Both these amino acid residues are basic amino acids and carry positive charges with them so histones are a set of positively charged basic proteins. These histones organise themselves to make a unit of eight molecules known as histone octamer. The negatively charged DNA is wrapped around the positively charged histone octamer forming a nonstructural unit known as nucleosome. One nucleosome contains 200 base pairs of DNA helix approximately. Nucleosomes are the repeating unit of chromatin which are threadlike stained bodies present in the nucleus. DNA helix of 200 bp wraps around the nucleosome by $1\frac{3}{4}$ turns.



NUCLEOSOME

OR

What is transcription? Explain the process of transcription in bacteria by labeled diagram.

Ans: The first process in the overall information transfer from DNA into protein structure is transcription. During this process a complementary mRNA strand is formed from the DNA strand. Transcription unit has primarily three regions-

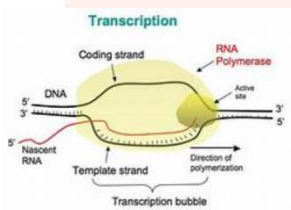
(i) Promoter – It marks the beginning of transcription. RNA polymerase binds here.

Structural gene – It is part of DNA which is primarily transcribed.

Terminator-It marks the end of transcription.

Transcription has three steps initiation, elongation and termination.

E coli RNA polymerase system consist of five different polypeptide chain refer to β , β' , β'' , ω , α , and has a molecular weight of 450000. The enzyme without the Sigma subunit is called core enzyme. The core enzyme is capable of transcribing DNA into RNA whereas the Sigma subunit helps in the recognition site of start signals on DNA molecule and direct DNA polymerase on in selecting the initiation site. Thereafter messenger RNA transcription begins at the initiation site of DNA. The first base transcribed for messenger RNA is purine. After the beginning of messenger RNA synthesis the Sigma unit dissociates. The dissociated Sigma unit can again combined with core enzyme to form RNA polymerase. Messenger RNA on DNA template takes place in in 5' to 3' direction so that successive nucleotides are attached at the 3'-OH end of growing mRNA. The termination of messenger RNA chain is brought about by certain termination signals on DNA. The termination signals are recognized by a protein factors called rho factor. A newly synthesized and RNA detached from DNA strand leaves the nucleus and becomes attached to ribosome.



TRANSCRIPTION IN BACTERIA

Explain in detail the role of restriction endonuclease information of Recombinant DNA. Draw a label diagram of E- coli cloning vector pBR 322. [3 + 1= 4]

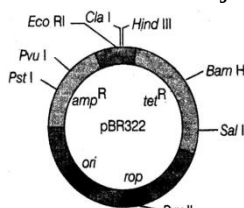
- Enzyme restriction endonuclease is used to identify and isolate the desired gene. These enzymes are molecular scissors which cut the DNA molecule at a specific site. Two enzymes from E. coli that were responsible for restricting the growth of bacteriophage were isolated in 1963, one of them added methyl group to DNA and the other cut DNA into segments. The later was called restriction endonuclease.
- Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the DNA. Palindrome in DNA is a sequence of base pairs that reads same on the two strands when orientation of reading is kept same.
- For example, the following sequences reads the same on the two strands in 5' → 3' direction as well as 3' → 5' direction. 5' — GAATTC — 3' 3' — CTTAAG — 5'

5' Mechanism

Restriction enzymes cut the strand of DNA little away from the centre of palindrome sides but between the same two bases on the opposite strand.

- This leaves single standard portions at each end .

- There are overhanging stretches called sticky Ends on each stand .These are name so because they form hydrogen bonds with their complementary cut counterpart .
- The stickiness of the ends facilitates the action of enzyme DNA ligase.
- Restriction endonucleases are used in genetic engineering to form a component molecules of DNA which are composed of DNA from different sources or genomes .
- These Sticky ends are complementary to each other when cut by same restriction enzyme therefore can be joined together end to end using DNA ligases.



E. Coli Cloning Vector

Or

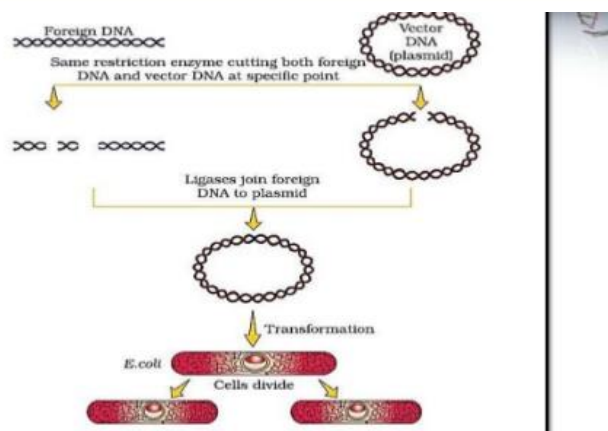
Define biotechnology. Mention all steps and explain only first two steps of Recombinant DNA technology in detail give diagrammatic representation of Recombinant DNA technology. [3 + 1= 4]

Ans: Biotechnology deals with the techniques of using living organisms or enzymes from organisms to produce products and processes useful to humans .Biotechnology refers to those processes which use a genetically modified organism to achieve the same on a large scale.

Recombinant DNA technology involves several steps in a specific sequence they are following-

- Isolation of genetic material (DNA)-** DNA is genetic material of most organisms. In order to cut the DNA with restriction enzymes it needs to be brought from free from other macromolecules such as RNA proteins polysaccharides and lipids. This is achieved by treating the bacterial cells plant and animal tissue with the enzyme such as lysosome cellulose and chitinase.
- Cutting of DNA at specific location-**In genetic engineering techniques usually a DNA molecule is broken by restriction enzyme at to desired places to isolate a specific DNA segment and then this segment is inserted into another DNA molecule at desired position. The joining of DNA involves several processes. The cut out gene of interest from the source DNA and the cut vector with space are mixed and ligase is added. The product thus obtained is called recombinant DNA.

Are



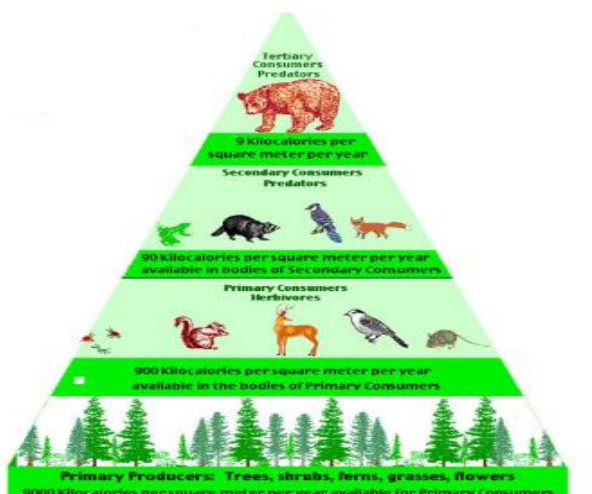
RECOMBINANT DNA TECHNOLOGY

What are ecological pyramids? What precaution should be kept in mind in calculating its different aspects? Draw the diagrams of pyramids of Grassland ecosystem and energy flow in ecosystem.

Ans: Ecological pyramids are diagrammatic representation showing the relationship between the Biomass, the number of energy content at different trophic level. They are considered as pyramid of biomass, pyramid of energy or pyramid of number.

Precaution to be kept in mind in calculating its different aspects are following-

Any calculations of energy content, Biomass or number has to include organism at the trophic level. New generalizations we make will be true if we take only a few individuals at any trophic level into account. Also a given organism may occupy more than one trophic level simultaneously. One must remember that the Trophic level represents a functional level not a species as such. A given species may occupy more than one trophic level in the same ecosystem at the same time. For example a sparrow is a primary consumer when it eats fruits piece and a secondary consumer when it eats insects and worms.



PYRAMID OF GRASSLAND

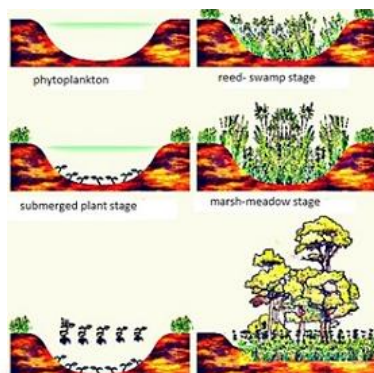
OR

What is ecological succession? Mention its different steps. Explain hydrach and xerach succession with diagrammatic representation. [1+1+1+1=4]

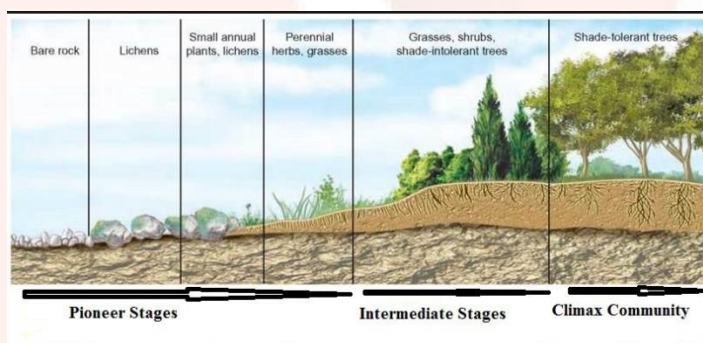
Ans: In course of time changes always keep on taking place in the community structure, Organization and the environment at a place in and this phenomena is called biotic or ecological succession. The process of succession always starts on a bare area and is completed through a number of steps. Major steps are as follows-

- Migration-The process of dispersal of seeds, spores and other structure of propagation of species to a bare area is called migration.
- Ecesis- Successful germination of propagules in a bare area is known as Ecesis.
- Colonization and aggregation-After ecesis the number of individuals of the species increase in number as a result of reproduction and they come close to each other. This process is called aggregation.
- Competition-Due to Limited sources species show both inter and intraspecific competition.
- Invasion-Various other types of plants try to establish in the space left by elimination of plants due to competition this is known as invasion
- Reaction-The newly arrived plants direct with the existing plants. This is known as reaction.
- Stabilization-Finally there after a stage in the process when the final terminal community becomes more or less stabilized for a long period of time it can maintain itself in equilibrium with the climate of the area. This process is not that stabilization.

Succession initiating in ponds, pools, lakes or elsewhere in water is called hydrarch. Hydrarch succession takes place in wetter area and succession series progress from hydric to the mesic condition. As against this xerarch succession takes place in dry areas and the series progress from xeric to mesic conditions. In both hydrarch and xerarch lead to medium water conditions neither too dry not too wet.



HYDRARCH SUCCESSION.



XERARCH SUCCESSION
